Lecture#23

1. What are the main methods of presentation of data?
2. Differentiate between line graph and sector graph.

Lecture#24

1. Calculate the mean , median and mode for the following data
   13, 18, 13, 14, 13, 16, 14, 21, 13.
   Answer: Mean=15, median=14, mode=13.

2. Calculate the range, class width, frequency and relative frequency for the following data
   36, 48, 54, 92, 57, 63, 66, 76, 66, 80.
   Answer: Range=56, class width=10.

Lecture#25

1. Calculate the arithmetic mean of the following grouped data

Distance	1-10	11-20	21-30	31-40
Number of persons	10	20	40	30

   Answer: Arithmetic mean=24.5

2. A die is thrown 20 times and the following scores are recorded
   6, 3, 2, 4, 5, 5, 6, 1, 3, 3, 5, 6, 6, 1, 3, 3, 5, 6, 6, 2.
   Find its arithmetic mean.
   Answer: Arithmetic mean=4.05

3. Calculate the frequency for the following data using MS Excel.
   80, 75, 90, 69, 83, 55, 79, 93, 58, 94, 89.

Lecture#26

1. Find the mean using the empirical relationship if median=15 and mode=22.
   Answer: Mean=11.5

2. Find the geometric mean and harmonic mean of 80, 85, 90, 95, 100.
   Answer: Geometric mean=89.72 and Harmonic mean=89.44

3. Check whether the following data set is symmetric, positively skewed or negatively skewed.
   27, 28, 30, 32, 34, 38, 41, 42, 44, 44, 46, 53, 56, 62.
   Answer: Positively skewed.

Lecture#27

1. Find all the quartiles for the following data set
   18, 20, 23, 20, 23, 27, 24, 23, 29.
   Answer: Q_1=20, Q_2=23 and Q_3=25.5

2. Calculate the mid hinge and quartile deviation for the following data set
   4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11.
   Answer: Mid hinge=9.75 and Quartile deviation=5.75

3. Find the range and midrange of the data set
   14, 10, 17, 5, 9, 20, 8, 24, 22.
   Answer: Range=19, midrange=14.5

Lecture#28

1. Calculate the standard deviation of first ten natural numbers (1-10).
   Answer: 2.872

2. Marks obtained by few students in a quiz are
   5, 6, 2, 3, 1, 7, 4, 8.
   Find the variance and coefficient of variance of the sample.
   Answer: Variance=5.25 and coefficient of variance=51%.

3. Team A scored the following number of goals in their last 20 matches:
   3, 0, 1, 5, 4, 3, 2, 6, 4, 2, 3, 3, 0, 7, 1, 1, 2, 3, 4, 6.
   Calculate the mean deviation about the mean.
   Answer: Mean deviation=1.5

Lecture#29

1. Calculate the correlation coefficient for the following data

X	1	2	4	6	7	10
y	9	6	4	12	8	3

   Answer: -0.268

2. Find the value of $cov(x,y)$, if $var(x)=100$, $var(y)=25$ and correlation coefficient $r=0.5$.
   Answer: $cov(x,y)=25$

Lecture#30

1. Find the least square regression line $y=a+bx$ for the following data

X	0	1	2	3	4
y	2	3	5	4	6

   Answer: $y=2.2+0.9x$

2. Calculate the intercept of regression line y=a+bx, if its slope is 0.55 associated with following data

X	3	5	7
y	6	4	2

   Answer: 1.25

3. Find the slope of the regression line $y=a+bx$ for the following data

X	2	10	4	8
y	6	7	3	4

   Answer: 0.15

Lecture # 31

Questions of Marks 3

Question:

Given $\sum XY = 400, \overline{X} = 5, \overline{Y} = 4, S_X = 4, S_Y = 3, n = 15$. Compute the coefficient of correlation between X and Y.

Solution:

$r = 0.55$

Question:

Determine whether two variables X and Y are correlated or uncorrelated by using the following table

X	6	3
Y	0	3

Solution:

$r = -1$

Question:

To calculate the intercept of the regression line we use equation $\bar{Y} = a + b\bar{X}$ when the slope of the line is 0.5. Determine whether Y intercept is positive or negative.

X	4	6	11	9
Y	5	10	9	12

Solution:

$a = 5.73$

Questions of Marks 5

Question:

Using the following table find Y intercept, slope and equation of regression line.

X	3	3	7	7
Y	2	9	6	9

Solution:

Intercept = 4

Slope = 0.5

Y = 4 + 0.5 X

Lecture # 32

Questions of Marks 2

Question:

If $Y = 7 + bX$, find the value of b when $Y = 12$ and $X = 3$

Solution:

$b = 1.67$

Question:

If $Y = a + 4X$, find the value of a when $Y = 15$and $X = 3$.

Solution:

$a = 3$

Questions of Marks 3

Question:

At any afternoon, the actual and trend values are 210 and 250 respectively; find the seasonal variation at the afternoon

Solution:

Seasonal variation = -40

Question:

If a point (5, 6) lie on the regression line $Y = -4 + bX$, find the slope b of the line

Solution:

$b = 2$

Questions of Marks 5

Question:

The estimated regression line of Y on X is

$Y = a + bX$, it passes through two points $A(7,9)$ and $B(0,0)$.Find the value of a and b.

Solution:

$a = 0$

b = 9/7

Question:

If $Y = 9 + bX$, find the value of b.

a) when Y = 5 and X = 2.

b) When Y = -7 and X = 4

Solution:

a)

b = -2

b)

b = - 4

Lecture # 33

Questions of Marks 3

Question:

From the data below:

Week no.	Actual sales	Forecast
3	7000	7500

Given that $\alpha = 0.3$, find the forecast for the 3rd week.

Solution:

Forecast week 4 = 7350

Questions of Marks 5

Question:

Find the centered average in the data below:

Quarter	Actual	Moving Average	Centered Average
1	72		
2	98		
3	79	88.75	
4	106	90.5	
1	79	96.5	
2	122	102	
3	101	111.25	
4	143	115	
1	94	119.75	
2	141	126.5	
3	128	130.75	
4	160	138.5	
1	125	139	
2	143	140.75	
3	135	147.5	
4	187		

Solution:

Quarter	Actual	Moving Average	Centered Average
1	72		
2	98		
3	79	88.75	89.625
4	106	90.5	----
1	79	96.5	99.25
2	122	102	----
3	101	111.25	113.125
4	143	115	----
1	94	119.75	----
2	141	126.5	----
3	128	130.75	----
4	160	138.5	138.75
1	125	139	----
2	143	140.75	----
3	135	147.5	
4	187		

Lecture 34

Question 1: Marks: 2

You have 3 hats and 4 shirts. A hat and a shirt can be chosen in ….. ways.

Solution:

Number of ways =12

Question 2: Marks: 3

How many different ways can you pick a team of 3 people from a group of 10.

Solution: 120.

Question 3: Marks: 3

How many different ways can you list 3 favorite desserts, in order, from a menu of 10.

Solution: 720

Question 4: Marks: 3

How many different arrangements are there of the letters of the word "***numbers***"?

Solution: 5,040

Question 5: Marks: 5

How many different 5-letter arrangements are there of the letters in the word digit ?

Solution:

60

Lecture 35

Question 1: Marks: 2

In how many ways can 3 different books be arranged on a shelf?

Solution:

Number of ways = 6 ways

Question:

There are 5 white balls and 4 black balls. In how many ways can we select 3 white and 2 black balls?

Solution:

$= {}_5C_3 \times {}_4C_2$

$= \frac{5!}{3!\,(5-3)!} \times \frac{4!}{2!(4-2)!}$

$= \frac{5\times4}{2!} \times \frac{4\times3}{2!}$

$= 10 \times 60$

= 60 ways

Question:

If a pair of dice is thrown then what is the probability of getting a total of 5 or 11?

Let X be ours discrete random variable which is the sum of the result of the two dices,them you know that:

P(X=11 or X=5) = P(X=11) + P(X=5) -P(X=11 and X=5)

But the sum of two dices can't be a same time 11 and 5 so P(X=11 and X=5) =0.

Them you know if you try all that the number of all sums possible are 36 = 6*6 and P(X=11) =2/36 = 1 /18 and P(X =5) = 4 /36 = 1/9 ,then:

P(X=11 or X =5) = 1/18 + 1/9 = 1/18 + 2/18 = 3/18 = **1/6**

Question 2: Marks: 3

Construct a sample space S, if you toss a coin TWO times and observe the sequence of heads (H) and tails (T) those appears.

Solution:

Required sample space is

$S = \{HH, HT, TH, TT\}$

Question 3: Marks: 5

A teacher must choose five monitors from a class of 12 students. How many different ways can the teacher choose the monitors?

Solution: 792

Lecture 36

Question 1: Marks: 2

Choose a number at random from 1 to 5. What is the probability of each outcome?

Solution: Probabilities: P(1) = 1/5 , P(2) = 1/5 , P(3) = 1/5 , P(4) = 1/5 , P(5) = 1/5

Question 2: Marks: 3

If you toss a die and observe the number of dots that appears on top face then write the event that an odd number AND a number exceeding 4 occurs.

Solution:

A = event that an odd number occur = $\{1,3,5\}$

B = event that an even number occur = $\{2,4,6\}$

C = event that a number greater than 4 occurs = $\{5,6\}$

Thus the event that an even number and a number exceeding 4 occurs is

$A \cap C = \{1,3,5\} \cap \{5,6\} = \{5\}$

Question 3: Marks: 5

Two dice are rolled, find the probability that the sum is

a) equal to 1

b) equal to 5

c) less than 10

Solution:

Solution:

The sample space S of two dice is shown below.

S = { (1,1),(1,2),(1,3),(1,4),(1,5),(1,6)
(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)
(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)
(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)
(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)
(6,1),(6,2),(6,3),(6,4),(6,5),(6,6) }

Let E be the event "sum equal to 1". There are no outcomes which correspond to a sum equal to 1, hence

P(E) = n(E) / n(S) = 0 / 36 = 0

b) Three possible ouctcomes give a sum equal to 5: E = {(1,4),(2,3),(3,2), (4,1)}, hence.

P(E) = n(E) / n(S) = 4 / 36 = 1 / 9

c) All possible ouctcomes, E = S, give a sum less than 10, hence.

P(E) = n(E) / n(S) = 30 / 36

Lecture 37

Question 1: Marks: 3

Find the number **n** of distinct permutations that can be formed from the letters of the word: APPLE.

Solution: 60

Question 2: Marks: 5

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur.

Solution:

1. Success = "A head is flipped on a single coin"
2. p = 0.5
3. q = 0.5
4. n = 10
5. x = 6

$P(x=6) = {}_{10}C_6 * 0.5^6 * 0.5^4 = 210 * 0.015625 * 0.0625 = 0.205078125$

Question 3: Marks: 5

Hospital records show that of patients suffering from a certain disease, 75% die of it. What is the probability that of 6 randomly selected patients, 4 will recover?

Solution:

This is a **binomial** distribution because there are only 2 outcomes (the patient dies, or does not).

The probability that 4 will recover = 0.0329595

Lecture 38

Question 1: Marks: 2

Write down formulas for the binomial probability mass function and cumulative binomial distribution.

Solution:

$$b(x;n,p)=\binom{n}{x}p^{x}(1-p)^{n-x}$$

$$B(x;n,p)=\sum_{y=0}^{n}\binom{n}{x}p^{n}(1-p)^{n-x}$$

Question 2: Marks: 5

Write down at least TWO properties of

1. Normal distribution.
2. Binomial Distribution.
3. Poisson distribution.

Lecture 39

Question 1: Marks: 3

Which type of distribution can better deal with the below mentioned cases?

- the number of deaths by car accidents
- birth defects and genetic mutations
- car accidents
- traffic flow and ideal gap distance
- number of typing errors on a page
- failure of a machine in one month

Question 2: Marks: 5

A company makes electric motors. The average number of defectives in 300 motors is $\mu = 3$. The probability an electric motor is defective is 0.01. What is the probability that a sample of 300 electric motors will contain exactly 5 defective motors?

Solution:

The probability of getting 5 defectives =0.10082

Lecture 40

Question 1: Marks: 5

Write down the formula and properties of a Normal Distribution.

Question 2: Marks: 3

Find proportion of bags which have weight in excess of 600 g.

Mean = 595.5g

St. Dev = 3 g

(For z=1.5 probability is 0.4332)

Solution:

z = (Value x - Mean value)/Standard deviation

z = (600- 595.5)/3 = 1.5

Given probability corresponding to z value is 0.4332

Proportion of bags which have weight in excess of 600 g.= 0.5 - 0.4332 = 0.0668